Question ID: b940952c

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	Easy

Question

On the basis of extensive calculations and models, astronomers in the 1990s predicted that the collision of two neutron stars or a neutron star and a black hole could release a massive burst of gamma rays in an event called a kilonova. This ______ was confirmed with observations in 2017.

Which choice completes the text with the most logical and precise word or phrase?

Answer

A. theory

B. evidence

C. constant

D. experiment

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text's discussion of a prediction about a kilonova. In this context, a "theory" is an explanation that is considered scientifically acceptable. The text states that astronomers predicted in the 1990s that a collision between a black hole and a neutron star or between two neutron stars could release a massive gamma ray burst called a kilonova, explaining that they determined this possibility based on their extensive work with existing data and simulations ("calculations and models"). In other words, the prediction was a theory—a well-supported explanation—that, as the text indicates, was later confirmed with observations in 2017.

Choice B is incorrect because the text indicates that it is the prediction made by astronomers in the 1990s that was confirmed in 2017, and a prediction of an event isn't "evidence," or proof, of that event's existence, even when the prediction is based on extensive study. Further, there would be no need for later confirmation of something that was already recognized as evidence. Choice C is incorrect because in this context, a "constant" is a situation or factor that doesn't change. The text indicates that it is the prediction made by astronomers in the 1990s that was confirmed in 2017, and there is no reason to describe the prediction as a constant because the text doesn't suggest that the prediction was completely unchanged over time—it addresses only the making of the prediction and its later confirmation. Choice D is incorrect because the text indicates that it is the prediction made by astronomers in the 1990s that was confirmed in 2017; although a prediction might be informed by an "experiment," or a controlled test, a prediction is an idea rather than a test.